EAN CUMMINGS

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PROFESSIONAL SUMMARY

M.S. Electrical & Computer Engineer specializing in Integrated Sensing, IoT ecosystems, and Embedded Systems. Proven track record of bridging hardware and software domains, from semiconductor cleanroom fabrication and LiDAR optics characterization to digital signal processing and full-stack application development.

EDUCATION

University of Iowa – College of Engineering **Iowa City, IA**

Master of Science in Electrical and Computer Engineering (M.S.) Anticipated: December 2026

- **Focus:** Integrated Sensing & Critical Infrastructure

Bachelor of Science in Biomedical Engineering (B.S.E.) May 2025

- **Focus Area:** Bioimaging; Image and Signal Processing, Embedded Systems
- **Study Abroad:** Dublin, Ireland (STEM & Irish Studies)

TECHNICAL SKILLS

- **Programming Languages:** Swift, Python, C++, Java, MATLAB, R, Alloy, Dafny
- **Hardware & Cleanroom Fabrication:** Photolithography, RF Magnetron Sputtering, Wet/Dry Etching, VASE, SEM, Circuit Design, Microcontrollers (ESP32/Arduino), Oscilloscopes, Precision TIG Welding (Titanium/Steel)
- **Software, Tools & Frameworks:** AutoCAD, Creo Parametric, ImageJ, Git, Firebase, InfluxDB, PSCAD, Power World

EXPERIENCE

LiDAR Systems Laboratory (NASA-Sponsored Research) | University of Iowa *Jan 2026 – May 2026*

- Designed, tested, and analyzed LiDAR system components spanning optical, electrical, and signal-processing subsystems for atmospheric research.
- Characterized laser beam divergence, footprint geometry, and front/back-surface Fresnel reflection at air-glass interfaces to model beam-steering optics and window calibration.
- Determined the angular field of view (FOV) and FWHM response of a detector-lens assembly, and validated range-dependent received-signal scaling.
- Analyzed photodiode transient response at 850 nm and 940 nm, extracting junction and diffusion capacitance from measured rise times using MATLAB.
- Designed and tuned transimpedance amplifier (TIA) feedback compensation networks to stabilize photodiode front-end circuits against capacitance-induced oscillation.

Microfabrication & Thin Film Laboratory | University of Iowa *Jan 2026 – May 2026*

- Executed end-to-end device fabrication process flows, including photoresist spin coating, bilayer resist lithography (LOR 5B / AZ1518), UV mask alignment/exposure, and wet/dry etching.
- Engineered a bilayer lift-off process achieving a controlled 1–2 µm undercut profile, verified via optical microscopy, to enable clean dielectric lift-off.
- Deposited 50 nm SiO_2 dielectric thin films via reactive RF magnetron sputtering, tuning Ar/O_2 gas flow ratios and process pressure to avoid target poisoning and ensure stoichiometric film growth.
- Performed anisotropic KOH wet etching of silicon using a sputtered SiO_2 hard mask; characterized etch depth and (111)-plane sidewall geometry via optical profilometry and SEM imaging.

- Characterized photoresist film thickness and uniformity using Variable Angle Spectroscopic Ellipsometry (VASE) fit to a Cauchy optical model.

PROJECTS

PuppyPOV – Wearable IoT Pet Health Monitor *IoT Course Project | Spring 2026*

- Hardware design, sensor integration, and edge-to-cloud backend for a wearable smart harness correlating canine biometrics (heart rate, HRV, SpO2, motion) with environmental context via event-triggered imaging.
- Integrated an Arduino Nano accelerometer, MAX30102 pulse oximeter, PiCam, and Adafruit Feather/OLED with a Raspberry Pi central hub over wired serial and I2C interfaces.
- Designed a custom DSP pipeline—including a 3rd-order Butterworth bandpass filter, median-filtered peak detection, and exponential smoothing—to extract reliable vital signs from motion-corrupted signals.
- Built a multithreaded, store-and-forward wireless architecture (HTTPS/TLS, offline queuing) streaming telemetry and images to Firebase and InfluxDB hubs.

InsulTrack – Smart Insulin Pen Case & iOS App *Capstone Project | 2024 – 2025*

- **Award:** 3rd Place, Hubert E. Storer Engineering Student Entrepreneurial Start-Up Award.
- Designed a thermally insulated, sensor-embedded smart case for diabetic management featuring automated dosage logging, GPS tracking, and environmental protection.
- Engineered a companion native iOS application using Swift and CoreBluetooth (BLE) for real-time data synchronization of blood glucose, insulin usage, exercise, and nutritional metrics.

Capsule Closet – Generative-AI Wardrobe Web Application *AI Project| Fall 2026*

- Web app deployed on Vercel that parses uploaded clothing photography to generate AI outfit try-on models via the Gemini API.
- Implemented client-side API key management features to enable secure, scalable, and cost-controlled API usage.

Electrocardiogram (EKG) Design & Build Health Project| Spring 2023

- Constructed a functional 3-lead EKG using multi-stage instrumentation amplifiers and analog filtering circuits.
- Interfaced hardware with an Arduino microcontroller for real-time heart rate plotting and automated atrial fibrillation (AFib) detection.

Custom Titanium and Steel Fabrication *Bike Project| 2024-2025*

- Engineered custom bicycle frames using AutoCAD for geometry optimization, tube-set profiling, and frame-jig alignment.
- Executed structural TIG welding on titanium and steel alloy elements, demonstrating strict atmospheric and material contamination control.

LEADERSHIP

Lead Student Ambassador *University of Iowa College of Engineering | 2024 – 2025*

- Trained, scheduled, and mentored a team of 18 student ambassadors, directing campus tours, information sessions, and panel discussions for prospective students and families.
- Organized and executed large-scale collegiate recruitment events and coordinated academic expert panels.

President *University of Iowa Triathlon Team | 2024 – 2025*

- Manage team operating budgets, corporate sponsorships, travel logistics, and weekly training schedules for club members.
- Organized and hosted Triowa (the University of Iowa indoor triathlon event) and competed at the USAT Collegiate Club National Championships.